

WaterSec AI Agent

Model Benchmark & Cost Analysis Report

Generated: 2026-05-10 01:31 | Data: Real IoT sensor data (158343 rows)

1 Groq Key Status

| Key   | Status    |
|-------|-----------|
| KEY_1 | Available |
| KEY_2 | Available |
| KEY_3 | Available |
| KEY_4 | Available |

Keys rotate automatically — rate-limited keys get a 60s cooldown before retry.

2 Overview

Compares **6** LLM candidates on **11** water domain prompts. **Real sensor data is injected into each prompt** so models answer with actual numbers. Scoring:  $S = 0.25 \cdot R + 0.30 \cdot C + 0.25 \cdot I + 0.20 \cdot D - 0.5 \cdot H$  (Relevance, Correctness, Insight, Domain 1–5; Hallucination 0–2).

3 Models Tested

| Model                | Type     | Speed    | In \$/1M | Out \$/1M | Monthly \$ |
|----------------------|----------|----------|----------|-----------|------------|
| Llama 3.1 8B (Groq)  | Hardware | 560 t/s  | \$0.050  | \$0.080   | \$0.15     |
| GPT-OSS 120B (Groq)  | Hardware | 1000 t/s | \$0.075  | \$0.300   | \$0.42     |
| Qwen3 32B (Groq)     | Hardware | 535 t/s  | \$0.290  | \$0.590   | \$1.01     |
| Llama 3.3 70B (Groq) | Hardware | 280 t/s  | \$0.590  | \$0.790   | \$1.63     |
| Mistral 7B (Local)   | Hardware | CPU      | \$0.000  | \$0.000   | \$0.00     |
| Qwen2.5 3B (Local)   | Hardware | CPU      | \$0.000  | \$0.000   | \$0.00     |

Monthly: 1,500 queries, avg 900 in + 700 out tokens. Hardware amortised 36mo/\$3,000.

4 Prompt-by-Prompt Results [real sensor data injected]

## Advanced Prompts

### [A1] Hard

Analyse the behavioural usage sequences detected in the residential CustomerC dataset within 5-minute windows. What do these sequences reveal about occupant behaviour and bathroom infrastructure? Provide specific water conservation recommendations backed by the occurrence numbers in the data.

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 1.1s

**\*\*Analysis of Behavioural Usage Sequences in Residential CustomerC Dataset\*\*** The data reveals a comprehensive picture of water usage patterns in a residential setting. The sequences indicate the most common transitions between water-using appliances within a 5-minute window. This analysis will focus on the top sequences and provide insights into occupant behavior and bathroom infrastructure. **\*\*Top Sequences:\*\*** 1. **\*\*Sink → Sink (30434 occurrences, 19% frequency)\*\***: This sequence suggests that occupants frequently use the sink for multiple purposes, such as washing hands, brushing teeth, or rinsing dishes. This behavior indicates a high frequency of handwashing and possibly inadequate water-saving practices. 2. **\*\*Tap → Sink (25821 occurrences, 16.1% frequency)\*\***: This sequence implies that occupants often turn on the tap before using the sink. This behavior may be due to a lack of awareness about water efficiency or inadequate faucet aerators, leading to unnecessary water flow. 3. **\*\*Tap → Flush (24428 occurrences, 15.2% frequency)\*\***: This sequence indicates that occupants frequently use the toilet after turning on the tap. This behavior may be due to a habit of washing hands before using the toilet or inadequate toilet flushing mechanisms. **\*\*Insights and Recommendations:\*\*** 1. **\*\*Handwashing habits:\*\*** The high frequency of sink usage suggests that occupants may not be adopting water-saving practices, such as turning off the tap while lathering hands or using a low-flow faucet... [truncated]

### GPT-OSS 120B (Groq) · 2.1s

## 1. What the 5-minute sequence data tell us about how the occupants use the bathroom suite | Rank | Sequence | Occurrences | % of all 5-min pairs | What it most likely means in home | |-----|-----|-----|-----|-----|-----| | 1 | **\*\*Sink → Sink\*\*** | **\*\*30 434\*\*** | **\*\*19 %\*\*** | Two separate sink uses within 5 min – classic “hand wash → hand wash”, “brush teeth → wash face”, or multiple people using the same sink one after the other. | | 2 | **\*\*Tap → Sink\*\*** | **\*\*25 821 | 16.1 %** | A tap is turned on (perhaps a tub or shower type faucet) and shortly after a sink event is recorded – could be a “fill the tub → rinse hands” or a “turn on the tap → adjust flow → start sink use”. | | 3 | **\*\*Tap → Flush\*\*** | **\*\*24 428 | 15.2 %** | A faucet is used and a toilet flush follows within 5 min – typical of a bathroom visit where the user washes hands then uses the toilet. | | 4 | **\*\*Flush → Sink\*\*** | **\*\*17 732 | 11 %** | Toilet flush first, then a sink use – e.g., “use toilet → wash hands”. | | 5 | **\*\*Sink → Flush\*\*** | **\*\*16 214 | 10.1 %** | Sink use first, then a flush – could be “brush teeth → use toilet” or “wash hands → flush after a diaper change”. | | 6 | **\*\*Flush → Flush\*\*** | **\*\*13 160 | 8.2 %** | Two flushes within 5 min – suggests either multiple occupants using the same toilet in quick succession (e.g., family members) or a single user performing a double flush (common with older single flush toilets). | | 7 | **\*\*Sink → Tap\*\*** | **\*\*12 483 | 7.8 %** | Sink use followed... [truncated]

### Qwen3 32B (Groq) · 2.3s

<think> Okay, let's start by looking at the data provided. The user wants an analysis of the behavioral usage sequences in the residential CustomerC dataset, focusing on 5-minute windows. They also want to know what these sequences reveal about occupant behavior and bathroom infrastructure, along with specific water conservation recommendations based on the data. First, I need to understand the sequences and their frequencies. The most common sequence is 'Sink → Sink' with 30,434 occurrences, which is 19% of the total. That's a high number. Then 'Tap → Sink' is next at 16.1%, followed by 'Tap → Flush' at 15.2%. The other sequences are all below 10%. I should consider what each sequence might indicate. For example, 'Sink → Sink' could mean people using the sink multiple times in quick succession, maybe for washing hands, brushing teeth, or filling containers. The high frequency suggests that this is a common activity pattern. Next, 'Tap → Sink' might indicate someone turning on the tap and then using the sink, perhaps for washing hands or face. 'Tap → Flush' could be related to using the tap and then flushing the toilet, maybe after washing hands. 'Flush → Sink' might be after using the toilet, then washing hands. The 'Sink → Flush' sequence could be someone using the sink and then flushing, maybe after brushing teeth and then using the toilet. 'Flush → Flush' might be consecutive flushes, which could indicate a problem with the toilet or someone flushing multiple times... [truncated]

### Llama 3.3 70B (Groq) · 1.8s

Analyzing the behavioural usage sequences in the residential CustomerC dataset reveals valuable insights into occupant behaviour and bathroom infrastructure. The most common sequence, 'Sink → Sink' (30434 times, 19% frequency), suggests that occupants frequently use the sink for consecutive tasks, such as washing hands, brushing teeth, or washing face, within a short time frame. The high occurrence of 'Tap → Sink' (25821 times, 16.1% frequency) and 'Sink → Tap' (12483 times, 7.8% frequency) sequences indicates that occupants often use the tap and sink in close proximity, possibly for tasks like filling a glass or washing vegetables. The presence of 'Tap → Flush' (24428 times, 15.2% frequency) and 'Sink → Flush' (16214 times, 10.1% frequency) sequences suggests that occupants use the toilet after using the tap or sink, which could be due to handwashing habits after using the bathroom. The 'Flush → Sink' (17732 times, 11% frequency) and 'Flush → Tap' (10680 times, 6.7% frequency) sequences imply that occupants often wash their hands after using the toilet, which is a good hygiene practice. To conserve water, the following recommendations are made based on the occurrence numbers: 1. **\*\*Install low-flow faucets\*\***: With 'Sink → Sink' being the most common sequence, installing low-flow faucets can significantly reduce water consumption. Assuming an average flow rate of 10 L/min, reducing the flow rate to 5 L/min can save approximately 152.17 L per day (30434 occurrences \\* 5-min... [truncated]

### Mistral 7B (Local) · 123.3s

Based on the analysis of the behavioral usage sequences in the CustomerC dataset, several insights about occupant behavior and bathroom infrastructure can be derived. 1. Handwashing and rinsing activities are the most frequent events, as indicated by the high

occurrences of 'Sink → Sink' (30434 times) and 'Tap → Sink' (25821 times). This suggests that hand hygiene is a priority for residents. 2. Toilet flushing is also a common event, with 'Tap → Flush' (24428 times) and 'Flush → Sink' (17732 times) being the second and fourth most frequent sequences. This indicates regular use of the bathroom facilities. 3. The sequence 'Sink → Flush' (16214 times) suggests that some residents may be washing their hands after using the toilet, which is a good hygiene practice. 4. The relatively low occurrences of 'Flush → Flush' (13160 times) and 'Tap → Tap' (9597 times) sequences might indicate that residents are not unnecessarily wasting water by repeatedly flushing the toilet or leaving the tap running. Based on these observations, here are some water conservation recommendations: 1. Encourage residents to practice water-saving techniques such as turning off the tap while washing hands or brushing teeth, which could reduce water usage in the 'Sink → Sink' and 'Tap → Sink' sequences. 2. Educate residents about the importance of not flushing the toilet unnecessarily, which could help decrease the 'Flush → Flush' sequence. 3. Promote dual-flush toilets or low-flow showerheads to furt... [truncated]

### Qwen2.5 3B (Local) · 74.8s

### Analysis of Residential CustomerC Dataset ##### Sequence Occurrences and Frequency: - \*\*Sequence\*\*: Sink → Sink (30,434 occurrences) - \*\*Sequence\*\*: Tap → Sink (25,821 occurrences) - \*\*Sequence\*\*: Tap → Flush (24,428 occurrences) - \*\*Sequence\*\*: Flush → Sink (17,732 occurrences) - \*\*Sequence\*\*: Sink → Flush (16,214 occurrences) - \*\*Sequence\*\*: Flush → Flush (13,160 occurrences) - \*\*Sequence\*\*: Sink → Tap (12,483 occurrences) - \*\*Sequence\*\*: Flush → Tap (10,680 occurrences) - \*\*Sequence\*\*: Tap → Tap (9,597 occurrences) ### Observations: The most common sequence detected is 'Sink → Sink' with 30,434 occurrences. This indicates that occupants frequently use the sink within a single 5-minute window after another event. The second most frequent sequence is 'Tap → Sink', which suggests that taps are often used to rinse or clean before using the sink. ### Potential Insights: 1. **Water Conservation in Bathroom Usage**: The high frequency of 'Sink → Sink' sequences implies that occupants may be rinsing their hands multiple times within a short period, indicating inefficient hand washing practices. 2. **Occupant Behavior**: The frequent use of taps (e.g., 'Tap → Sink', 'Tap → Flush') suggests that occupants might not be turning off the tap while brushing teeth or using the toilet, leading to unnecessary water waste. ### Recommendations for Water Conservation: 1. **Hand Washing Efficiency**: - Encourage occupants to turn off the faucet while rinsing their hands. - Install... [truncated]

### [A2] Hard

*Design a demand-based hot water heating schedule for the gym based on the real hourly sensor data. Identify peak hours, off-peak windows, and recommend specific pre-heating start times. Estimate expected energy savings as a percentage. Account for the fact that hot water usage is ~4x higher than col...*

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 1.2s

Based on the provided data, I'll analyze the hot water usage patterns and design a demand-based hot water heating schedule for the gym. **Peak Hours Analysis**: The top 5 peak hours for hot water usage are: 1. 21:00 - 18,362.1 L 2. 20:00 - 15,040.7 L 3. 14:00 - 9,755.0 L 4. 09:00 - 9,482.0 L 5. 19:00 - 9,466.2 L These peak hours indicate that the gym experiences high hot water demand during late evening (21:00, 20:00), mid-afternoon (14:00), and early morning (09:00) hours. **Off-Peak Windows**: To identify off-peak windows, I'll analyze the hourly hot water usage data. Assuming a 24-hour period, I'll categorize hours with low hot water demand as off-peak. Off-peak windows: \* 02:00 - 06:00 (low usage, average 1,500 L/hour) \* 07:00 - 08:00 (low usage, average 2,500 L/hour) \* 10:00 - 13:00 (low usage, average 2,500 L/hour) \* 15:00 - 18:00 (low usage, average 2,500 L/hour) \* 22:00 - 23:00 (low usage, average 1,500 L/hour) **Pre-Heating Start Times**: To optimize hot water heating, I recommend pre-heating the system during off-peak windows. Based on the flow rate and hot water demand, I suggest the following pre-heating start times: \* 02:00 - 06:00: Pre-heat the system 30 minutes before the first peak hour (07:00) \* 07:00 - 08:00: Pre-heat the system 30 minutes before the 09:00 peak hour \* 10:00 - 13:00: Pre-heat the system 30 minutes before the 14:00 peak hour \* 15:00 - 18:00: Pre-heat the system 30 minutes before the 19:00 peak hour \* 22:00 - 23:00: Pre-heat the system ... [truncated]

### GPT-OSS 120B (Groq) · 2.1s

**Demand-based hot water heating schedule for the gym** (All calculations are drawn directly from the sensor data you supplied – no external numbers are invented.) --- ## 1. Identify the true demand peaks | Rank | Hour (24 h) | Hot water volume (all time) | |-----|-----|-----| | 1 | 21:00 | 18 362.1 L | | 2 | 20:00 | 15 040.7 L | | 3 | 14:00 | 9 755.0 L | | 4 | 09:00 | 9 482.0 L | | 5 | 19:00 | 9 466.2 L | These five hours account for the largest share of hot water demand and therefore define the **peak hour windows** that must be covered by a ready-to-use hot water buffer. --- ## 2. Define off-peak windows The remaining 19 hours of the day show far lower cumulative usage. Using the flow rate distribution (5th

### Qwen3 32B (Groq) · 3.1s

<think> Okay, let's tackle this problem step by step. The user wants a demand-based hot water heating schedule for a gym based on sensor data. They also need peak hours identified, off-peak windows, pre-heating start times, and energy savings estimates. Plus, they mention that hot water usage is four times higher than cold, which I need to consider. First, I'll look at the provided data. The top 5 peak hours for hot water are 21:00, 20:00, 14:00, 09:00, and 19:00. The highest is at 21:00 with 18,362.1 liters. The flow rate data shows a median of 3.95 L/min, which might help in estimating usage patterns. Since hot water is 4x higher than cold, I need to adjust the total demand. But the data given is already for hot water, so maybe that's already factored in? Wait, the problem states that hot water usage is ~4x higher than cold. So perhaps the total water usage is a combination, but the sensor data here is specifically for hot. Hmm, maybe the 4x is a ratio between hot and cold. So when calculating total demand, I need to consider that hot is 4x cold. But the data provided is only for hot. Maybe the user wants the heating schedule to account for the higher hot water usage compared to cold, but since the data is already hot, perhaps the 4x is a red herring? Or maybe it's about energy consumption, where heating hot water uses more energy. Wait, the problem says "hot water usage is ~4x higher than cold water usage." So total water usage is hot + cold, but hot

is 4x cold. So if th... [truncated]

### Llama 3.3 70B (Groq) · 1.8s

To design a demand-based hot water heating schedule, we'll analyze the provided data. The top 5 peak hours for hot water usage are: 1. 21:00 with 18,362.1 L 2. 20:00 with 15,040.7 L 3. 14:00 with 9,755.0 L 4. 09:00 with 9,482.0 L 5. 19:00 with 9,466.2 L These hours indicate the periods of highest hot water demand. To optimize energy usage, we'll identify off-peak windows and recommend pre-heating start times. Off-peak windows can be assumed to be the hours not listed in the top 5 peak hours, as the data only provides the top 5 peak hours. Recommended pre-heating start times: - For the 09:00 peak, pre-heating can start at 07:00 (2 hours before the peak). - For the 14:00 peak, pre-heating can start at 12:00 (2 hours before the peak). - For the 19:00, 20:00, and 21:00 peaks, pre-heating can start at 17:00 (2 hours before the first peak). Expected energy savings: Since hot water usage is ~4x higher than cold water usage, optimizing hot water heating can lead to significant energy savings. By pre-heating only during the recommended periods, the gym can avoid heating water during off-peak hours. Assuming the heating system operates at a constant efficiency, the energy savings can be estimated based on the reduction in heating time. Given the top 5 peak hours account for the majority of hot water usage, and assuming the gym operates 24/7, the pre-heating schedule can reduce heating time by approximately 60-70% (14-16 hours of non-peak time out of 24 hours). However, since the... [truncated]

### Mistral 7B (Local) · 185.4s

Based on the provided data, the top 5 peak hours for hot water usage at the gym are: 1. 21:00 - 18,362.1 L (highest consumption) 2. 20:00 - 15,040.7 L 3. 14:00 - 9,755.0 L 4. 09:00 - 9,482.0 L 5. 19:00 - 9,466.2 L Since hot water usage is approximately four times higher than cold water usage, we can assume that the peak hours for total water consumption are similar to the ones provided above. To design a demand-based hot water heating schedule, we will focus on the peak hours and identify off-peak windows. The peak hours will require preheating, while off-peak windows can have lower temperature water or no heating at all. Peak Hours: - 21:00 - 23:59 (7,684.1 L on average) - 20:00 - 21:59 (14,585.4 L on average) - 14:00 - 15:59 (9,755.0 L on average) - 09:00 - 10:59 (9,482.0 L on average) - 19:00 - 20:59 (9,466.2 L on average) Off-Peak Windows: - 00:00 - 08:59 - 11:00 - 13:59 - 16:00 - 18:59 - 22:00 - 06:59 Recommended Pre-heating Start Times: - For the 21:00 peak hour, preheat hot water starting from 19:00. - For the 20:00 peak hour, preheat hot water starting from 18:30. - For the 14:00 peak hour, preheat hot water starting from 13:30. - For the 09:00 peak hour, preheat hot water starting from 07:30. - For the 19:00 peak hour, preheat hot water starting from 18:00. By focusing on the peak hours and adjusting the pre-heating start times, we can optimize the hot water heating schedule to better match demand. This will result in energy savings as the water heater won't b... [truncated]

### Gwen2.5 3B (Local) · 90.4s

To design a demand-based hot water heating schedule for the gym based on the provided data, we need to identify peak hours and off-peak windows, recommend specific pre-heating start times, and estimate expected energy savings. #### Peak Hours and Off-Peak Windows Based on the top 5 peak hours identified from the sensor data: 1. \*\*21:00\*\* — 18,362.1 L 2. \*\*20:00\*\* — 15,040.7 L 3. \*\*14:00\*\* — 9,755.0 L 4. \*\*09:00\*\* — 9,482.0 L 5. \*\*19:00\*\* — 9,466.2 L #### Pre-Heating Start Times To determine pre-heating start times, we need to consider the flow rate and how much hot water is needed in each peak hour. ##### Flow Rate Analysis: The median flow rate is 3.95 L/min, with a range of 0.51 – 15.06 L/min. For simplicity, let's use the median flow rate for calculations. - \*\*21:00\*\* — 18,362.1 L - \*\*20:00\*\* — 15,040.7 L - \*\*14:00\*\* — 9,755.0 L - \*\*09:00\*\* — 9,482.0 L - \*\*19:00\*\* — 9,466.2 L #### Pre-Heating Start Times Calculation: For each peak hour: - Calculate the total flow rate required for that hour. - Determine how much hot water needs to be pre-heated before the start of the peak hour. ##### Example Calculation for \*\*21:00\*\* (18,362.1 L): - Total flow rate = 3.95 L/min - Duration = 1 hour = 60 minutes Total litres needed:  $\text{Litres} = 3.95 \times 60 = 237 \text{ L}$  Since the gym needs to have enough hot water for a total of 18,362.1 L in this hour:  $\text{Pre-heating required} = \frac{18,362.1 - 237}{3.95} \approx 4,500 \text{ L}$ ... [truncated]

### [A3] Hard

*CustomerB has a single aggregated sensor covering 2 WCs (man/woman), each with a central sink, 2 flushes, 2 flexibles, and 1 wudu sink — 9+ fixtures on 1 sensor. What are the specific risks of this architecture for leak detection and anomaly isolation? Redesign the sensor layout: state the minimum n...*

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 2.1s

Based on the provided data, I'll analyze the risks associated with the current architecture and propose a redesign for leak detection and anomaly isolation. **Current Architecture Risks:** 1. **False Positives:** With 9+ fixtures on a single sensor, it's challenging to pinpoint the source of a leak or anomaly. A leak in one fixture might trigger a false positive for another fixture, leading to unnecessary maintenance and potential water waste. 2. **Masking of Leaks:** If a leak occurs in one fixture, the sensor may not detect it immediately, especially if the leak is small. This can lead to undetected leaks, resulting in wasted water and increased water bills. 3. **Difficulty in Isolating Anomalies:** With multiple fixtures on a single sensor, it's hard to determine which fixture is causing an anomaly, making it challenging to isolate and repair the issue. **Redesign Proposal:** To mitigate these risks, I recommend the following sensor layout: 1. **WC Area:** Place a separate sensor for each WC (man and woman) to monitor their respective fixtures (central sink, 2 flushes, and 2 flexibles). This will provide accurate data for each WC and help identify leaks or anomalies specific to each area. 2. **Sink Area:** Install a separate sensor for the central sink to monitor its usage independently. 3. **Wudu Sink:** Place a separate sensor for the wudu sink to monitor its usage separately, as it's a distinct area with its own fixtures. 4. **Flushes:** Install separate sensors fo... [truncated]

### GPT-OSS 120B (Groq) · 2.0s

### 1. Risks of a **single sensor, 9 fixture aggregation** | Risk | Why it matters for leak detection / anomaly isolation | How it shows up in the data you supplied | |-----|-----|-----| | **No fixture level granularity** | The sensor only reports the **total** volume (214 731.3 L) and the **total** flow time for the whole bathroom suite. If a leak appears on one of the two WC flushes, the system cannot tell which WC is responsible – it only sees an “extra” L/min that could be caused by any of the nine outlets. | A slow leak on a WC ( $\approx 0.5 \text{ L min}^{-1}$ ) will be masked by normal sink use ( $\approx 2 \text{ L min}^{-1}$ ) and will appear as a modest, unexplained rise in the overall average flow rate. | | **High false positive rate** | Normal usage patterns for a bathroom are highly variable (e.g., a burst of water when both sinks are used simultaneously). With one aggregated stream the algorithm must treat any deviation from the long-term average as a potential leak, leading to many alerts that turn out to be legitimate usage spikes. | The 214 731.3 L over ~7 months averages to  $\approx 0.42 \text{ L min}^{-1}$  ( $\approx 25 \text{ L h}^{-1}$ ). A short-term “peak” of  $1 \text{ L min}^{-1}$  (common when both sinks run) would be flagged as an anomaly even though it is perfectly normal. | | **Inability to isolate the source** | When an alert fires, maintenance staff have to shut down **all** fixtures to locate the leak, increasing downtime and cost. | The data set only ... [truncated]

### Qwen3 32B (Grok) · 3.2s

<think> Okay, let's tackle this problem. The user is asking about the risks of having a single sensor for multiple fixtures in Customer B's setup and wants a redesign with the minimum number of sensors. First, I need to understand the setup. Customer B has 9+ fixtures on one sensor. The fixtures include two WCs (men's and women's), each with a central sink, two flushes, two flexibles, and one wudu sink. So that's a lot of different points where water is used. The main issue with a single sensor is that it aggregates all the data. If there's a leak, the sensor can't tell which fixture is leaking. For example, if the sensor detects a sudden increase in water usage, it's impossible to know if it's the sink, the toilet, or the wudu sink. This makes anomaly isolation difficult. Also, different fixtures have different usage patterns. A toilet might have short bursts of high flow, while a sink might have longer, lower flows. A single sensor might not capture these nuances, leading to false positives or missed leaks. Another risk is that the aggregated data might mask smaller leaks. If multiple fixtures are slightly leaking, the total might not be significant enough to trigger an alert. Plus, maintenance teams can't target specific areas, leading to inefficient repairs and higher water waste. Now, for the redesign. The goal is to minimize the number of sensors while improving detection and isolation. Let's break down the fixtures: - Two WCs (men and women): Each has a central s... [truncated]

### Llama 3.3 70B (Grok) · 1.8s

Based on the provided data, CustomerB has a single aggregated sensor covering 9+ fixtures, which poses several risks for leak detection and anomaly isolation: 1. **Leak detection difficulty**: With a single sensor covering multiple fixtures, it may be challenging to identify the source of a leak. If a leak occurs, the sensor will detect increased water usage, but it will be difficult to determine which fixture is causing the leak. 2. **Anomaly isolation**: If an anomaly is detected, such as unusual water usage patterns, it will be hard to isolate the specific fixture or area causing the anomaly. 3. **False positives**: A single sensor may generate false positives, where a legitimate water usage event is flagged as an anomaly, due to the combined water usage of multiple fixtures. To mitigate these risks, I recommend redesigning the sensor layout to include multiple sensors. The minimum number of sensors needed would be 3-4, depending on the specific layout and fixture grouping. Here's a possible sensor placement strategy: 1. **WC area sensor**: Place one sensor to cover both WCs, including their respective sinks. This will allow for monitoring of water usage patterns specific to the WCs. 2. **Wudu sink sensor**: Install a separate sensor for the wudu sink, as it may have distinct water usage patterns compared to the WCs. 3. **Flexible fixture sensor**: Place a sensor to cover the 2 flexible fixtures, which may include sinks, showers, or other water-using appliances. 4. **Op...** [truncated]

### Mistral 7B (Local) · 79.5s

Given the current architecture of CustomerB's sensor setup with 9+ fixtures on a single sensor, there are several risks associated with leak detection and anomaly isolation: 1. **Single point of failure**: If this sensor fails or malfunctions, it could lead to an inability to monitor water usage for all the fixtures, making it difficult to detect leaks or anomalies. 2. **Inadequate granularity**: With so many fixtures on a single sensor, it might be challenging to isolate anomalies or pinpoint the source of a leak, as the data from all fixtures is combined. This could result in delayed detection and repair of leaks. 3. **Leak localization**: In case of a leak, it would be difficult to determine which specific fixture is affected without additional information or further investigation. To address these issues, I recommend redesigning the sensor layout as follows: 1. **Minimum number of sensors needed**: At least 3 sensors are required to cover all the fixtures effectively. This will help reduce the risk of a single point of failure and improve granularity in data collection. 2. **Sensor placement**: - Sensor A: Place it near the central sink, covering the sink and the two flushes. This sensor will monitor water usage for washing hands, doing dishes, and flushing the toilets. - Sensor B: Position this sensor near the wudu sink to monitor water usage specifically for wudu purposes. - Sensor C: Install this sensor near the flexible hoses to monitor any potential le... [truncated]

### Qwen2.5 3B (Local) · 53.2s

### Analysis of CustomerB's Sensor Architecture ##### Risks: **Leak Detection**: The single sensor covering all fixtures (2 WCs, 2 flushes, 2 flexibles, and 1 wudu sink) makes it difficult to pinpoint leaks. If a leak occurs in any fixture, the total consumption will be affected, making it hard to isolate which specific fixture is leaking. **Anomaly Isolation**: Isolating anomalies (abnormal usage patterns or spikes) becomes challenging due to the aggregated data from all fixtures on one sensor. For example, if only one fixture experiences a significant increase in usage, this might not be detected as an anomaly because it's part of the total consumption for that sensor. ### Redesigning Sensor Layout ##### Minimum Number of Sensors Needed: To improve leak detection and anomaly isolation, we need to distribute sensors across different fixtures. Here's how: 1. **2 WCs**: Each WC should have its own sensor. - CustomerB has 2 WCs, so we need 2 additional sensors. 2. **2 Flushes**: These can share a single sensor as they are likely used together and their consumption patterns might be similar. - No additional sensors needed for flushes. 3. **2 Flexibles**: Each flexible should have its own sensor. - CustomerB has 2 flexibles, so we need 2 additional sensors. 4. **1 Wudu Sink**: This can share a single sensor with the closest fixture (WC or flexible) as it is likely used in conjunction with other fixtures. - No additional sensors needed for wudu sink. #####



Total... [truncated]

**[A4] Expert**

Rank the 4 monitored sites by leak detection risk (highest to lowest): gym showers (8 sensors, hot+cold), office toilets (4 aggregated bloc sensors), sanitary bloc (1 aggregated sensor), residential (7 granular sensors). For each site: (1) justify the ranking, (2) set an anomaly threshold in litres ...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the provided data, I'll make an assumption about the average session duration for each site. Since the actual session duration is not provided, I'll use the average consumption to estimate the session duration. 1. **\*\*Gym showers (8 sensors, hot+cold)\*\***: - Average consumption: 6.11 L - Estimated session duration: 6.11 L / 0.5 L/min (a reasonable flow rate for showers) = 12.22 minutes or 734 seconds - Justification: High leak detection risk due to multiple sensors and high water usage. - Anomaly threshold: 15 L (3 times the average consumption) - Alert action chain: - Initial alert: WaterSec sends a notification to the facilities manager within 1 minute via email and SMS. - Follow-up alert: If the anomaly persists for 30 minutes, the facilities manager is notified again, and the building management team is alerted via email. 2. **\*\*Office toilets (4 aggregated bloc sensors)\*\***: - Average consumption: 2.19 L - Estimated session duration: 2.19 L / 0.2 L/min (a reasonable flow rate for toilets) = 10.95 minutes or 659 seconds - Justification: Moderate leak detection risk due to aggregated sensors and moderate water usage. - Anomaly threshold: 5 L (2.3 times the average consumption) - Alert action chain: - Initial alert: WaterSec sends a notification to the facilities manager within 1 minute via email and SMS. - Follow-up alert: If the anomaly persists for 15 minutes, the facilities manager is notified again, and the build... [truncated]

GPT-OSS 120B (Groq) · 2.0s

**\*\*Leak■detection■risk ranking (highest → lowest)\*\*** | Rank | Site | Why the risk is highest / lowest |  
 |-----|-----|-----| | **\*\*1\*\*** | **\*\*Sanitary bloc (1 aggregated sensor)\*\*** | Only a single, block■level sensor reports the  
 \*total\* volume for the whole bloc. A small, continuous leak (e.g., a dripping pipe) can be hidden by normal usage and will only appear  
 as

## Qwen3 32B (Groq) · 3.4s

<think> Okay, let's tackle this problem step by step. The user wants me to rank four monitored sites by their leak detection risk from highest to lowest. The sites are gym showers, office toilets, sanitary bloc, and residential. Each has different sensor setups: gym has 8 sensors (hot and cold), office has 4 aggregated bloc sensors, sanitary bloc has 1 aggregated sensor, and residential has 7 granular sensors. First, I need to understand what factors contribute to leak detection risk. More sensors might mean better monitoring, but aggregated sensors could miss individual leaks. The average consumption per session is given for gym showers (6.11 L) and customerC (2.19 L). Wait, customerC isn't one of the sites listed. Maybe that's a typo? The user mentioned four sites: gym showers, office toilets, sanitary bloc, and residential. So maybe customerC is a typo for residential? The average for residential is 2.19 L per session. That makes sense because residential usage is typically lower than gym showers. Leak risk is higher where there's potential for unnoticed continuous flow. Gym showers might have higher usage per session, but with 8 sensors, individual leaks could be detected. Office toilets with 4 aggregated sensors might miss individual leaks if one of the sensors in a bloc is faulty. Sanitary bloc with 1 sensor is a big risk because a single point of failure could mean a leak goes undetected. Residential has 7 granular sensors, which is good for individual monitoring, bu... [truncated]

## Llama 3.3 70B (Grog) · 1.9s

Based on the provided data, I will rank the 4 monitored sites by leak detection risk from highest to lowest: 1. **\*\*Gym showers\*\***: - Justification: The gym showers have the highest average water consumption per session (6.11 L), which suggests a higher potential for water loss in case of a leak. Additionally, the presence of 8 sensors (hot and cold) provides more opportunities for detecting anomalies. - Anomaly threshold: 15 L per event (approximately 2.5 times the average session consumption) - Alert action chain: - Immediate notification (within 1 minute) to the gym's maintenance team via SMS and email - Escalation to the facility manager within 15 minutes if the issue is not acknowledged - Automatic generation of a work order for investigation and repair within 30 minutes 2. **\*\*Residential\*\***: - Justification: The residential site has 7 granular sensors, which provides detailed usage data and increases the likelihood of detecting leaks. However, the average water consumption per event is not provided, making it difficult to assess the risk accurately. - Anomaly threshold: Unable to set a specific threshold due to lack of data on average residential water consumption per event. A general threshold of 10 L per event could be considered, but this may need to be adjusted based on actual usage patterns. - Alert action chain: - Notification to the resident via mobile app and email within 5 minutes - Escalation to the residential manage... [truncated]

## Mistral 7B (Local) · 112.0s

Based on the data provided, let's rank the sites by leak detection risk from highest to lowest: 1. Gym Showers (8 sensors, hot+cold): The gym showers consume significantly more water per event compared to other sites, with an average of 6.11 L per session. This high consumption suggests a higher likelihood of leaks or inefficient usage. 2. Office Toilets (4 aggregated bloc sensors): Although the office toilets have lower average consumption than gym showers, they still consume more water per event compared to other sites (average of 2.19 L per session). The aggregated nature of these sensors might mask smaller leaks, but their higher consumption still indicates a relatively high leak detection risk. 3. Sanitary Bloc (1 aggregated sensor): With only one sensor for the entire sanitary bloc, it is more difficult to detect small leaks or inefficient usage compared to sites with multiple granular sensors. However, since its average consumption is lower than office toilets and gym showers, it ranks third in terms of leak detection risk. 4. Residential (7 granular sensors): The residential site has the lowest average consumption per event (not specified), and it also has more granular sensors

which can help detect smaller leaks or inefficient usage. Therefore, it ranks lowest in terms of leak detection risk. Anomaly Threshold: - Gym Showers: 10 L per event (2.5 times the average consumption) - Office Toilets: 3 L per event (1.4 times the average consumption) - Sanitary Bloc: 1.... [truncated]

### Qwen2.5 3B (Local) · 67.5s

Given the data provided and the absence of detailed analysis from the `module 'tools'`, I will base my assessment on the average water usage per event for each site, assuming that the gym's sensor provides an accurate representation of their usage. Here's how we can rank them by leak detection risk (highest to lowest): ### 1. Gym Showers - **Avg Usage**: 6.11 L - **Anomaly Threshold**: Since this is a high-volume area with multiple sensors, I will set the anomaly threshold at 20% above the average usage: -  $\text{Threshold} = 6.11 \text{ L} \times (1 + 0.2) = 7.332 \text{ L}$  - **Alert Action Chain**: - **Who is Notified**: Facility Manager and Maintenance Team - **Notification Timeframe**: Immediate alert via SMS, email, and in-app notification - **Channel**: SMS, Email, In-App Notification ### 2. Office Toilets (4 Aggregated Sensors) - **Avg Usage**: 2.19 L - **Anomaly Threshold**: For office toilets, I will set the threshold at 30% above the average usage: -  $\text{Threshold} = 2.19 \text{ L} \times (1 + 0.3) = 2.847 \text{ L}$  - **Alert Action Chain**: - **Who is Notified**: Facility Manager and Maintenance Team - **Notification Timeframe**: Immediate alert via SMS, email, and in-app notification - **Channel**: SMS, Email, In-App Notification ### 3. Sanitary Bloc (1 Aggregated Sensor) - **Avg Usage**: Since no specific data is provided for this site, I will use the office toilets' average as a proxy: - Avg Usage: 2.19 L - **Anomaly Threshold**: For sanitary blocs, set at 40%... [truncated]

### [A5] Expert

*Residential Flush 2 recorded 2796.63 L in a single event (z-score: 11.75). A normal flush is 6 L. Flow rate during this event was 6 L/min. Calculate step by step: 1. Exact duration of the event in minutes and hours 2. Water cost at 0.003 EUR/L 3. CO2 equivalent at 0.298 kg CO2 per m³ of water...*

### Llama 3.1 8B (Groq) · 1.3s

To calculate the duration of the event, we need to use the flow rate and the total consumption. 1. Given flow rate = 6 L/min, and total consumption = 2796.63 L, we can calculate the duration of the event as follows: Duration = Total consumption / Flow rate Duration = 2796.63 L / 6 L/min Duration = 466.44 min To convert the duration to hours, we divide by 60: Duration (hours) = 466.44 min / 60 Duration (hours) = 7.774 hours 2. To calculate the water cost, we multiply the total consumption by the cost per liter: Water cost = Total consumption \* Cost per liter Water cost = 2796.63 L \* 0.003 EUR/L Water cost = 8.38989 EUR 3. To calculate the CO2 equivalent, we first need to convert the total consumption from liters to cubic meters (1 m³ = 1000 L): Total consumption (m³) = 2796.63 L / 1000 Total consumption (m³) = 2.79663 m³ Then, we multiply the total consumption in cubic meters by the CO2 equivalent per cubic meter: CO2 equivalent = Total consumption (m³) \* CO2 equivalent per m³ CO2 equivalent = 2.79663 m³ \* 0.298 kg CO2/m³ CO2 equivalent = 0.832 kg CO2 Ranking the three most likely root causes by probability: 1. **Clogged drain or pipe**: This is the most likely cause, given the extremely high consumption and z-score of 11.75. A clogged drain or pipe would restrict the water flow, causing the flow rate to decrease, and the water to accumulate, resulting in the high consumption. Probability: High 2. **Faulty or stuck valve**: A faulty or stuck valve could also caus... [truncated]

### GPT-OSS 120B (Groq) · 2.1s

**\*1. Exact duration of the event\*** - Total volume recorded: **2796.63 L** - Reported flow rate: **6 L / min**  $\text{Duration (min)} = \frac{\text{Volume}}{\text{Flow rate}} = \frac{2796.63 \text{ L}}{6 \text{ L/min}} = 466.105 \text{ min}$   $\text{Duration (h)} = \frac{466.105 \text{ min}}{60} = 7.7684167 \text{ h}$  **Result:** - **466.105 minutes** ( $\approx 7 \text{ hours } 46 \text{ minutes}$ ) - **7.77 hours** (rounded to two decimals) --- **\*2. Water cost (0.003 EUR / L)**  $\text{Cost} = 2796.63 \text{ L} \times 0.003 \frac{\text{EUR}}{\text{L}} = 8.38989 \text{ EUR}$  **Result:** **€8.39** --- **\*3. CO2 equivalent emissions (0.298 kg CO2 / m³)** Convert litres to cubic metres:  $2796.63 \text{ L} = \frac{2796.63}{1000} \text{ m}^3 = 2.79663 \text{ m}^3$   $\text{CO2} = 2.79663 \text{ m}^3 \times 0.298 \frac{\text{kg CO2}}{\text{m}^3} = 0.83339574 \text{ kg CO2}$  **Result:** **0.83 kg CO2** --- ## Ranking of the three most likely root cause categories | Rank | Likely cause | Why it fits the data (probability) | ---|---|---| **\*1\*** | **Faulty/stuck valve or continuous flow malfunction** | • The volume ( $\approx 2.8 \text{ m}^3$ ) is  $> 460 \times$  the normal 6 L flush. <br>• The flow rate stayed at a plausible 6 L/min **over 7 hours**, which is typical of a valve that failed to close rather than a brief user action. <br>

### Qwen3 32B (Groq) · 2.0s

<think> Okay, let's tackle this problem step by step. First, I need to calculate the duration of the event. The user mentioned that the flow rate was 6 L/min, and the total consumption was 2796.63 L. To find the duration, I can divide the total liters by the flow rate. So 2796.63 L divided by 6 L/min gives me the time in minutes. Let me do that calculation:  $2796.63 / 6 = 466.105$  minutes. Then, converting that to hours, I divide by 60.  $466.105 / 60$  is approximately 7.768 hours. So the duration is about 7 hours and 46 minutes. Next, calculating the water cost. The cost is 0.003 EUR per liter. Multiplying 2796.63 L by 0.003 EUR/L gives 8.38989 EUR. Rounding that to two decimal places, it's approximately 8.39 EUR. For the CO2 equivalent, the rate is 0.298 kg CO2 per cubic meter. Since 1 cubic meter is 1000 liters, I need to convert 2796.63 L to cubic meters by dividing by 1000. That's 2.79663 m³. Multiplying by 0.298 kg/m³ gives 0.8334 kg CO2. Rounded to three decimal places, that's 0.833 kg CO2. Now, ranking the root causes. The z-score is 11.75, which is way above normal. A normal flush is 6 L, but this event used over 2700 L. The flow rate is 6 L/min, which is normal for a toilet, but the duration is extremely long. So possible causes: 1) A toilet cistern leak where the valve is stuck open, causing continuous flow. 2) A faulty fill valve that doesn't shut off after the tank fills. 3) A broken float mechanism leading to overfilling. The first two are more likely since they i... [truncated]

### Llama 3.3 70B (Groq) · 2.3s

To calculate the required values, we'll follow the steps provided: 1. **Exact duration of the event in minutes and hours**: - Given: consumption\_L = 2796.63 L, flow rate = 6 L/min - Duration in minutes = Total litres / Flow rate per minute =  $2796.63 \text{ L} / 6 \text{ L/min} =$

466.105 minutes - To convert minutes to hours, we divide by 60:  $466.105 \text{ minutes} / 60 = 7.76842 \text{ hours}$  2. **\*\*Water cost at 0.003 EUR/L\*\***: - Water cost = Total litres \* Cost per litre =  $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.38989 \text{ EUR}$  3. **\*\*CO2 equivalent at 0.298 kg CO2 per m³ of water\*\***: - First, convert litres to cubic meters ( $1 \text{ m}^3 = 1000 \text{ L}$ ):  $2796.63 \text{ L} / 1000 = 2.79663 \text{ m}^3$  - CO2 equivalent = Volume in m³ \* CO2 per m³ =  $2.79663 \text{ m}^3 * 0.298 \text{ kg CO2/m}^3 = 0.833 \text{ kg CO2}$  Now, to rank the three most likely root causes by probability and explain each, considering the provided data: 1. **\*\*Continuous Flow Due to Faulty Valve or Sensor (High Probability)\*\***: - The extremely high z-score (11.75) and the large volume of water used (2796.63 L) compared to a normal flush (6 L) suggest a malfunction. A continuous flow over 7.77 hours without interruption is highly indicative of a mechanical or sensor issue preventing the valve from closing properly. 2. **\*\*Incorrect Installation or Maintenance (Medium-High Probability)\*\***: - The prolonged duration and excessive water usage could also result from incorrect installation or recent maintenance that inadvertently caused the valve to remain open. This scenario is less likely than ... [truncated]

### Mistral 7B (Local) · 113.3s

1. To calculate the duration of the event in minutes, we first convert the total litres used (consumption\_L) to cubic meters (m³). Since there are 1000 L in a m³, we get:  $\text{Duration in m}^3 = \text{consumption\_L} / 1000 = 2.79663 \text{ m}^3$  Now, let's find the volume of water in cubic meters per second (m³/s) by dividing the total litres used by the session duration in seconds:  $\text{Flow rate in m}^3/\text{s} = \text{consumption\_L} / \text{session\_duration\_s} = 2.79663 \text{ L} / 60 \text{ s} = 0.04661 \text{ m}^3/\text{s}$  Since there are 60 seconds in a minute and 3600 seconds in an hour, we can now calculate the duration of the event in minutes:  $\text{Duration\_min} = \text{session\_duration\_s} / 60 = 60 \text{ s} / 60 \text{ min/s} = 1 \text{ min}$  And in hours:  $\text{Duration\_h} = \text{duration\_min} / 60 = 1 \text{ min} / 60 \text{ min/h} = 0.0167 \text{ h}$  (approximately) 2. To calculate the water cost, we multiply the total litres used by the price per liter:  $\text{Water cost} = \text{consumption\_L} * \text{price\_per\_liter} = 2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.38999 \text{ EUR}$  3. To calculate the CO2 equivalent, we first find the volume of water in cubic meters (m³) and then multiply it by the CO2 emission factor:  $\text{CO2\_equivalent} = \text{consumption\_L} * \text{CO2\_emission\_factor} = 2796.63 \text{ L} * 0.298 \text{ kg CO2/m}^3 = 831.45 \text{ kg CO2}$  (approximately) Ranking the three most likely root causes by probability: 1. Leak or excessive usage: The consumption of water during this event is significantly higher than a normal flush, indicating that there might be a leak or excessive usage. 2. Faulty fixture or appliance: A faulty toilet, showerhead,... [truncated]

### Gwen2.5 3B (Local) · 57.0s

Let's break down the problem step-by-step: ### Step 1: Exact Duration of the Event in Minutes and Hours Given: - Total litres used (consumption\_L) = 2796.63 L - Flow rate during this event (session\_duration\_s) = 600 seconds/min or 10 L/s To find the duration in minutes, we use:  $\text{Duration (minutes)} = \frac{\text{Total litres used}}{\text{Flow rate (L/min)}} = \frac{2796.63}{10} = 279.663 \text{ minutes}$  To convert this to hours, we divide by 60:  $\text{Duration (hours)} = \frac{279.663}{60} \approx 4.661 \text{ hours}$  ### Step 2: Water Cost at 0.003 EUR/L Given the total litres used and the cost per litre:  $\text{Water cost (EUR)} = 2796.63 \times 0.003 = 8.400 \text{ EUR}$  ### Step 3: CO2 Equivalent at 0.298 kg CO2 per m³ of water First, convert litres to cubic meters ( $1 \text{ L} = 0.001 \text{ m}^3$ ):  $\text{Total volume in m}^3 = 2796.63 \times 0.001 = 2.79663 \text{ m}^3$  Now calculate the CO2 emissions:  $\text{CO2 emissions (kg)} = 2.79663 \times 0.298 \approx 0.834 \text{ kg CO}_2$  ### Ranking of Root Causes by Probability 1. **\*\*High Flow Rate\*\***: The event had a flow rate of 10 L/s, which is significantly higher than the normal flush (6 L/min or 1 L/s). This suggests that there might be an issue with the plumbing system, such as a leak or malfunctioning fixture. 2. **\*\*Excessive Duration\*\***: An event lasting approximately 4.66 hours for what should have been a brief flush indicates potential misuse of water resources. It could also suggest issues like fa... [truncated]

## [A6] Expert

A water utility wants to use WaterSec to meet ISO 24512 water service management standards. Based on the 4 customer profiles and their sensor granularity, identify: (1) which ISO 24512 KPIs can be computed directly from this sensor data and how, (2) which KPIs require additional data sources — name ...

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 1.5s

**\*\*1. ISO 24512 KPIs computable directly from WaterSec sensor data:\*\*** Based on the provided data, the following ISO 24512 KPIs can be computed directly from WaterSec sensor data: - **\*\*KPI 1.1.1: Total water consumption\*\*** (already provided in the data) - For each customer: total litres used over the monitoring period (e.g., customerA: 105,712.5 L) - For all customers: total litres used over the monitoring period (e.g., total: 544,205.5 L) - **\*\*KPI 1.1.2: Average water consumption\*\*** (already provided in the data) - For each customer: average litres used per day (e.g., customerA: 604.1 L/day) - For all customers: average litres used per day (e.g., total: 744.1 L/day) - **\*\*KPI 1.2.1: Water consumption by time of day\*\*** (not directly available, but can be inferred from session\_duration\_s) - This KPI requires additional data sources, such as time stamps for each usage event. - **\*\*KPI 1.2.2: Water consumption by day of the week\*\*** (not directly available, but can be inferred from session\_duration\_s) - This KPI requires additional data sources, such as time stamps for each usage event. - **\*\*KPI 1.3.1: Water consumption by meter\*\*** (already provided in the data) - For each customer: total litres used over the monitoring period (e.g., customerA: 105,712.5 L) - **\*\*KPI 1.3.2: Meter accuracy\*\*** (not directly available, but can be inferred from the data) - This KPI requires additional data sources, such as calibration records for each meter. - **\*\*KPI 2.1.1: Water loss\*\*** (not di... [truncated])

### GPT-OSS 120B (Groq) · 2.1s

## 1. ISO 24512 KPIs that can be **\*\*calculated directly\*\*** from the WaterSec sensor feed | ISO 24512 KPI (or sub-KPI) | What the sensor data gives you | How to compute it with the numbers you already have |  
|-----|-----|-----|  
| **\*\*Total water consumption (volume supplied)\*\*** | `total L` per customer profile | Sum the four "total L" values → 105 712.5 L + 214 731.3 L + 162 082.1 L + 162 679.6 L = **\*\*645 205.5 L\*\*** for the reporting period. | | **\*\*Average Daily Consumption (L / day)\*\*** | `L/day` per profile (already derived) | Use the



"L/day" column directly, or recompute as `total L ÷ days measured`. Example for Customer A:  $105\,712.5\text{ L} \div 175\text{ d} = 604.1\text{ L/day}$  (matches the supplied figure). | **Consumption per customer segment** | Separate totals for A, B, C, gym | Report each segment's total volume and its share of the overall volume (e.g., Customer A =  $105\,712.5\text{ L} \div 645\,205.5\text{ L} \approx 16.4\%$ ). | **Water Use Intensity (L / customer / day)** | "L/day" per profile | The "L/day" column already expresses intensity per individual (or per "

### Qwen3 32B (Groq) · 3.3s

<think> Okay, let's tackle this problem step by step. The user wants to use WaterSec's sensor data to meet ISO 24512 standards. First, I need to recall what ISO 24512 entails. From what I remember, it's about water service management, focusing on performance metrics like water supply reliability, efficiency, and customer satisfaction. The key is to identify which KPIs can be directly calculated from the given data and which need more information. Looking at the data provided: four customers with their daily water usage and total over different periods. The data includes consumption in liters per day and total liters over days. The first part of the question asks which KPIs can be computed directly. Let's think about possible KPIs. Water consumption per customer is straightforward—each customer's daily average is given. Total water supplied per period can be calculated by summing up the totals for each customer. For example, customerA's total is 105,712.5 L over 175 days. But wait, the data already provides the total liters for each customer over their respective periods. So maybe the total water supplied per period is just the sum of all customers' totals. That would be  $105,712.5 + 214,731.3 + 162,082.1 + 162,679.6 =$  let me add those up. Let's see: 105k + 214k is 319k, plus 162k is 481k, plus another 162k is 643k total liters. So that's a direct KPI. Next, average daily consumption per customer is already given in the data (e.g., customerA: 604.1 L/day). So that's another... [truncated]

### Llama 3.3 70B (Groq) · 2.6s

Based on the provided WaterSec sensor data, I'll address the three parts of your question: **1. ISO 24512 KPIs Computable from Sensor Data** The following ISO 24512 KPIs can be computed directly from the sensor data: - **Per capita water consumption**: Although the data doesn't provide the number of people in each customer profile, if we assume the number of inhabitants or users for each profile, we can calculate per capita consumption by dividing the total consumption by the number of people. - **Average daily water consumption**: Already provided in the data (e.g., 604.1 L/day for customerA). - **Total water consumption**: Also provided in the data (e.g., 105,712.5 L for customerA over 175 days). - **Water consumption per day**: Directly available from the data. To compute these KPIs, we use the formulas: - Average daily water consumption = Total consumption / Number of days. - Total water consumption = Given in the data. - Water consumption per day = Given in the data. **2. KPIs Requiring Additional Data Sources** The following KPIs require additional data sources: - **Water loss**: Requires data on the total amount of water supplied to the area and the amount of non-revenue water (e.g., leaks, unauthorized use). - **Continuity of service**: Needs data on the frequency and duration of water supply interruptions. - **Customer satisfaction**: Requires survey data or feedback from customers regarding the water service. - **Water quality**: Needs data from water quality t... [truncated]

### Mistral 7B (Local) · 113.6s

(1) ISO 24512 KPIs that can be computed directly from this sensor data are: - **Average Daily Consumption (ADC)**: This can be calculated by dividing the total consumption for each customer by the number of days they were monitored. For example, ADC for customerA is 604.1 L/day. - **Total Consumption (TC)**: This is already provided in the data as the total liters used over the monitoring period for each customer. (2) ISO 24512 KPIs that require additional data sources are: - **Peak Daily Consumption (PDC)**: To calculate PDC, we need to know the maximum daily consumption recorded during the monitoring period. This requires analyzing the session\_duration\_s and consumption\_L data for each day. - **Minimum Daily Consumption (MDC)**: Similar to PDC, MDC can be found by identifying the minimum daily consumption during the monitoring period. Again, this requires analyzing the session\_duration\_s and consumption\_L data for each day. - **Billing Period Consumption (BPC)**: To calculate BPC, we need information about the billing cycle length for each customer. This is not provided in the current data. - **Consumption Percentiles**: To calculate consumption percentiles, we would need more granular data on individual usage events throughout the day, which is not currently available. (3) Proposed Monthly Compliance Report Structure: Title: WaterSec Monthly Compliance Report Sections: 1. Executive Summary - Total Consumption for each customer - Average Daily Consumption for each customer 2. Deta... [truncated]

### Qwen2.5 3B (Local) · 83.9s

### Analysis of ISO 24512 KPIs and Compliance Report Structure ##### (1) Directly Computed KPIs from Sensor Data: **Customer A**: - **Water Usage Rate (L/min)**: - Total consumption = 105,712.5 L - Total days = 175 days - Daily average usage = 604.1 L/day - Assuming a day has 1440 minutes:  $\frac{604.1}{1440} \approx 0.417$  L/min **Customer B**: - **Water Usage Rate (L/min)**: - Total consumption = 214,731.3 L - Total days = 214 days - Daily average usage = 1003.4 L/day - Assuming a day has 1440 minutes:  $\frac{1003.4}{1440} \approx 0.697$  L/min **Customer C**: - **Water Usage Rate (L/min)**: - Total consumption = 162,082.1 L - Total days = 628 days - Daily average usage = 258.1 L/day - Assuming a day has 1440 minutes:  $\frac{258.1}{1440} \approx 0.179$  L/min **Gym**: - **Water Usage Rate (L/min)**: - Total consumption = 162,679.6 L - Total days = 746 days - Daily average usage = 218.1 L/day - Assuming a day has 1440 minutes:  $\frac{218.1}{1440} \approx 0.152$  L/min **KPIs that can be computed directly from the sensor data include**: - **Water Usage Rate (L/min)**: Calculated based on total daily usage and average days per month. - **Daily Average Consumption**: Total consumption divided by total days. ##### (2) KPIs Requiring Additional Data Sources: 1. **Total Monthly Water Usage**: Requires additional data to calculate monthly totals from the daily sensor readings. 2. **Water Efficiency Metrics**: Such as water effic... [truncated]

## Standard Prompts

### [S1] Easy

What is the total water consumption at the gym last month? The gym has 4 shower cabins, each with a hot and cold sensor (8 sensors total). Analyse the monthly trend and state the most recent full month total in litres.

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 3.0s

To find the total water consumption at the gym last month, we need to identify the most recent full month in the data. The most recent full month is 2026-04-30. The total water consumption for 2026-04-30 is 10,898.8 L. To analyze the monthly trend, we can calculate the total water consumption for each month: - 2024: The data only includes one month (April), with a total consumption of 1,280 L. - 2025: The total consumption for 2025 is the sum of the consumption for each month: - January: 7,990.48 L - February: 6,440.59 L - March: 3,847.6 L - April: 6,100.08 L - May: 5,724.53 L - June: 1,311.83 L - July: 3,075.66 L - August: 9,697.42 L - September: 9,954.82 L - October: 14,118.6 L - November: 12,331.9 L - December: 12,136.4 L The total consumption for 2025 is 92,528.51 L. - 2026: The total consumption for 2026 is the sum of the consumption for each month: - January: 15,387.7 L - February: 14,540.2 L - March: 7,825.56 L - April: 10,898.8 L The total consumption for 2026 (up to April) is 48,552.06 L. The monthly trend shows an increase in water consumption over the years, with the highest consumption in 2026.

### GPT-OSS 120B (Groq) · 2.1s

**\*\*Most recent full month total (April 2026)\*\* - \*\*Consumption:\*\* 10 898.8 L\*\* - \*\*Average flow rate for the month:\*\* 4.715 L /min\*\***  
--- **## Monthly Trend (last 12 months) | Consumption (L) | Flow rate (L /min) |**  
2025-06-30 | 1 311.83 | 5.598 | 2025-07-31 | 3 075.66 | 4.811 | 2025-08-31 | 9 697.42 | 5.364 | 2025-09-30 | 9 954.82 | 5.219 | 2025-10-31 | 14 118.6 | 4.764 | 2025-11-30 | 12 331.9 | 4.452 | 2025-12-31 | 12 136.4 | 4.311 | 2026-01-31 | 15 387.7 | 6.141 | 2026-02-28 | 14 540.2 | 7.545 | 2026-03-31 | 7 825.56 | 5.301 | 2026-04-30 | 10 898.8 | 4.715 |  
2026-05-31\* | 2 914.66 | 5.658 | \*May 2026 is only partially complete (today is 10 May), so it is **\*\*not\*\*** a full month figure. **### What the numbers tell us | Observation | Detail |**  
**\*\*Overall growth\*\*** | From June 2025 (1 312 L) to January 2026 (15 388 L) the gym's water use rose **\*\*1 070 %\*\*** – a clear upward trajectory as membership and shower usage increased. | **\*\*Peak month\*\*** | **\*\*January 2026\*\*** recorded the highest consumption (15 387.7 L) with the highest average flow rate (6.141 L/min). | **\*\*Recent dip\*\*** | February 2026 stayed high (14 540 L) but March 2026 dropped sharply to 7 825.6 L, likely reflecting a short-term reduction in gym attendance (e.g., holiday period). | **\*\*Recovery\*\*** | April 2026 rebounded to 10 898.8 L, a **\*\*39 %\*\*** increase over March and **\*\*30 %\*\*** above the ... [truncated]

### Qwen3 32B (Groq) · 3.5s

<think> Okay, let's see. The user is asking for the total water consumption at the gym last month. The gym has 4 shower cabins, each with hot and cold sensors, making 8 sensors total. They want the monthly trend analyzed and the most recent full month's total in liters. First, I need to look at the data provided. The data starts from 2024-04-30 up to 2026-05-31. The last date in the data is 2026-05-31 with 2914.66 liters. But the user is asking for the most recent full month. Wait, the current date isn't specified, but assuming today's date is after 2026-05-31, the most recent full month would be May 2026. However, looking at the data, the last entry is May 2026 with 2914.66 liters. But wait, the next month is June 2026, but there's no data for June. Wait, the data goes up to 2026-05-31. So the most recent full month in the data is May 2026. But the user might be asking for the latest month available. Let me check the dates again. Looking at the periods: the last row is 2026-05-31 with 2914.66 L. The next period would be June 2026, but it's not present. So the most recent full month in the data is May 2026. However, if the current date is, say, June 2026, then May would be the last full month. But since the data stops at May 2026, that's the latest. So the answer should be 2914.66 liters for May 2026. Wait, but the user says "last month" which could be relative to the current date. If the current date is June 2026, then last month is May. But if the data ends in May 2026, t... [truncated]

### Llama 3.3 70B (Groq) · 1.8s

The most recent full month total water consumption at the gym is for the month of April 2026, with a total consumption of 10898.8 litres. To determine the monthly trend, we can observe the total consumption over the past few months: - 2026-01: 15387.7 L - 2026-02: 14540.2 L - 2026-03: 7825.56 L - 2026-04: 10898.8 L - 2026-05: 2914.66 L (partial month, not considered for trend analysis) The trend shows a decrease in consumption from January to March, followed by an increase in April. However, the consumption in May is significantly lower, but this data is for a partial month and may not be representative of the full month's consumption.

### Mistral 7B (Local) · 140.8s

To find the total water consumption at the gym last month (February 2026), we need to look for the data corresponding to the period 2026-02-01 to 2026-02-29, as there is no data for February 29th due to it being a leap year. However, in the provided data, there's no record for February 2026. Therefore, I cannot determine the total water consumption at the gym last month based on the given data. Regarding the monthly trend analysis, we can observe that the water consumption varies significantly from month to month. The lowest consumption was recorded in July 2024 (849.47 L) and the highest was recorded in January 2026 (15387.7 L). The flow rate also fluctuates, with the lowest being 1.28 L/min in April 2024 and the highest being 7.545 L/min in February 2026. However, without the data for February 2026, I cannot provide the most recent full month total water consumption in litres.

### Qwen2.5 3B (Local) · 51.4s

To determine the total water consumption at the gym last month, we need to identify the most recent full month total in litres from the provided data. The most recent full month is April 2026 with a total of 10898.8 litres consumed. Therefore, the total water consumption at the gym last month (April) was **\*\*10898.8 L\*\***.

**[S2] Easy**

Compare hot vs cold water usage across all 4 gym shower cabins. Calculate the hot-to-cold ratio per cabin and identify which cabin uses the most hot water overall. What does this suggest operationally?

✓ Real sensor data injected into this prompt

**Llama 3.1 8B (Groq) · 0.7s**

Based on the provided data, here's the comparison of hot vs cold water usage across all 4 gym shower cabins: **Hot-to-Cold Ratio per Cabin:** - Cabin 1: 4.3x (hot water usage is 4.3 times higher than cold water usage) - Cabin 2: 5.6x (hot water usage is 5.6 times higher than cold water usage) - Cabin 3: 4.0x (hot water usage is 4.0 times higher than cold water usage) - Cabin 4: 2.8x (hot water usage is 2.8 times higher than cold water usage) **Cabin with the Most Hot Water Usage:** - Cabin 2 uses the most hot water overall, with a total of 39,222.8 L. **Operational Implications:** - The high hot-to-cold ratio in Cabin 2 suggests that users in this cabin are more likely to use hot water for longer periods or in greater quantities compared to the other cabins. - This could be due to various factors such as user preferences, the type of activities being performed in the cabin, or the design of the shower facilities. - To optimize water usage and reduce energy consumption, the gym could consider installing temperature controls or timers to encourage users to use hot water more efficiently. - Additionally, the gym may want to consider providing education or signage to raise awareness about the importance of water conservation and encourage users to adopt more sustainable showering habits.

**GPT-OSS 120B (Groq) · 2.1s**

**Hot vs Cold Water Use by Cabin (litres)**

| Cabin | Hot water (L) | Cold water (L) | Hot ÷ Cold Ratio |
|-------|---------------|----------------|------------------|
| 1     | 45,386.4      | 10,664.1       | 4.3              |
| 2     | 39,222.8      | 7,039.8        | 5.6              |
| 3     | 30,778.5      | 7,729.5        | 4.0              |
| 4     | 16,164.9      | 5,693.5        | 2.8              |

**Which cabin uses the most hot water?** - Cabin 1 consumes the greatest amount of hot water: 45,386.4 L. **Overall picture:** - **Total hot water** across the four cabins: 131,552.6 L - **Total cold water** across the four cabins: 31,126.9 L - **Combined hot to cold ratio** for the gym:  $131,552.6 \div 31,126.9 \approx 4.2$  (hot water is more than four times the cold water volume). **Operational implications:** 1. Cabin 1 is the biggest hot water draw. - Likely sees the highest shower frequency, longest shower durations, or a larger user base. - It may be a priority for targeted efficiency measures (e.g., low flow showerheads, temperature mix controls, or user education signage). 2. **Hot water demand dominates overall consumption.** - The heating system must be sized to meet a peak hot water load that is roughly four times the cold water load. - Opportunities exist to reduce energy use by: \* Installing thermostatic mixing valves that limit the hot water proportion while still delivering comfortable temperatures. \* Adding heat recovery

**Qwen3 32B (Groq) · 3.3s**

<think> Okay, let's tackle this problem step by step. The user wants me to compare hot and cold water usage across four gym shower cabins, calculate the hot-to-cold ratio for each, identify which cabin uses the most hot water overall, and then discuss what that suggests operationally. First, I need to look at the data provided for each cabin. The data already includes the hot and cold water usage in liters and the hot-to-cold ratio. Wait, the user provided the ratios as part of the data. But maybe they want me to verify those ratios? Let me check. For example, Cabin 1 has Hot: 45,386.4 L and Cold: 10,664.1 L. The ratio is  $45,386.4 / 10,664.1 \approx 4.256$ , which rounds to 4.3x as given. So the provided ratios are correct. So maybe the user just wants me to confirm that and then proceed. Next, the task is to calculate the hot-to-cold ratio per cabin. Since the ratios are already provided, perhaps the user wants me to present them again but maybe in a different format or with more context. But the main point is to identify which cabin uses the most hot water. Looking at the hot water usage numbers: Cabin 1: 45,386.4 L Cabin 2: 39,222.8 L Cabin 3: 30,778.5 L Cabin 4: 16,164.9 L Clearly, Cabin 1 has the highest hot water usage. So that's straightforward. Now, the operational implications. The highest hot water usage in Cabin 1 could suggest a few things. Maybe it's the most frequently used cabin, or the showers there are used for longer durations, leading to higher hot water consu... [truncated]

**Llama 3.3 70B (Groq) · 1.3s**

To compare hot vs cold water usage across all 4 gym shower cabins, we can analyze the provided data. The total hot water usage across all cabins is:  $45,386.4 \text{ L} + 39,222.8 \text{ L} + 30,778.5 \text{ L} + 16,164.9 \text{ L} = 131,552.6 \text{ L}$  The total cold water usage across all cabins is:  $10,664.1 \text{ L} + 7,039.8 \text{ L} + 7,729.5 \text{ L} + 5,693.5 \text{ L} = 31,126.9 \text{ L}$  The overall hot-to-cold ratio is:  $131,552.6 \text{ L (hot)} / 31,126.9 \text{ L (cold)} = 4.2x$  The hot-to-cold ratio per cabin is already provided in the data: - Cabin 1: 4.3x - Cabin 2: 5.6x - Cabin 3: 4.0x - Cabin 4: 2.8x Cabin 2 has the highest hot-to-cold ratio (5.6x), indicating it uses the most hot water relative to cold water. Cabin 1 uses the most hot water overall, with 45,386.4 L. This suggests operationally that Cabin 1 may have the highest demand for hot water, possibly due to factors such as user preference or shower duration. Cabin 2's high hot-to-cold ratio may indicate a potential for energy-saving opportunities by optimizing the hot water temperature or encouraging users to use more cold water. Cabin 4's lower hot-to-cold ratio (2.8x) may indicate a different user behavior or showering pattern compared to the other cabins.

**Mistral 7B (Local) · 119.0s**

To compare hot vs cold water usage across all four gym shower cabins, we can calculate the hot-to-cold ratio for each cabin by dividing the total litres of hot water used by the total litres of cold water used. 1. Cabin 1: Hot-to-cold ratio =  $45,386.4 \text{ L} / 10,664.1 \text{ L} = 4.3x$  2. Cabin 2: Hot-to-cold ratio =  $39,222.8 \text{ L} / 7,039.8 \text{ L} = 5.6x$  3. Cabin 3: Hot-to-cold ratio =  $30,778.5 \text{ L} / 7,729.5 \text{ L} = 4.0x$  4. Cabin 4: Hot-to-cold ratio =  $16,164.9 \text{ L} / 5,693.5 \text{ L} = 2.8x$  Now, let's identify which cabin uses the most hot water overall by finding the highest hot-to-cold ratio. The cabin with the highest ratio uses the most hot water: 1. Cabin 1: Ratio = 4.3x 2. Cabin 2: Ratio = 5.6x (uses the most hot water) 3. Cabin 3: Ratio = 4.0x 4. Cabin 4: Ratio = 2.8x Operational implications of this finding could include: 1. Higher energy consumption for heating water in Cabin 2, as it uses more hot water compared to the other cabins. 2. Potential for water and energy savings by optimizing hot water usage in Cabin 1, Cabin 3, and Cabin 4, aiming to match or reduce their hot-to-cold ratios towards that of Cabin 2. 3. Investigating the reasons behind the differences in hot water usage between cabins, such as user preferences, shower

settings, or maintenance practices. 4. Implementing strategies to encourage users to use cold water more often, if appropriate for their needs, to reduce energy consumption and costs associated with heating water.

### Qwen2.5 3B (Local) · 64.7s

To analyze the hot-to-cold water usage ratio per cabin and identify which cabin uses the most hot water overall, we need to calculate the total hot water used in each cabin and then compare these totals. #### Cabin 1: - Hot Water Used: 45,386.4 L - Cold Water Used: 10,664.1 L Total Hot Water Usage for Cabin 1: \*\*45,386.4 L\*\* #### Cabin 2: - Hot Water Used: 39,222.8 L - Cold Water Used: 7,039.8 L Total Hot Water Usage for Cabin 2: \*\*39,222.8 L\*\* #### Cabin 3: - Hot Water Used: 30,778.5 L - Cold Water Used: 7,729.5 L Total Hot Water Usage for Cabin 3: \*\*30,778.5 L\*\* #### Cabin 4: - Hot Water Used: 16,164.9 L - Cold Water Used: 5,693.5 L Total Hot Water Usage for Cabin 4: \*\*16,164.9 L\*\* Now we compare the total hot water usage across all cabins: - \*\*Cabin 1:\*\* 45,386.4 L - \*\*Cabin 2:\*\* 39,222.8 L - \*\*Cabin 3:\*\* 30,778.5 L - \*\*Cabin 4:\*\* 16,164.9 L \*\*Cabin 1\*\* uses the most hot water overall with a total of \*\*45,386.4 L\*\*. #### Hot-to-Cold Ratio Analysis: We already have the ratios calculated for each cabin: - \*\*Cabin 1:\*\* Hot: 4.3x - \*\*Cabin 2:\*\* Hot: 5.6x - \*\*Cabin 3:\*\* Hot: 4.0x - \*\*Cabin 4:\*\* Hot: 2.8x #### Operational Implications: The highest hot-to-cold ratio is observed in Cabin 1, indicating that this cabin uses significantly more hot water relative to cold water compared to the other cabins. This suggests that Cabin 1 might have a higher demand for hot water or possibly less efficient water heating systems, leading to higher energy consumption and potentially hi... [truncated]

### [S3] Medium

Identify and explain the top anomalies in the residential CustomerC dataset. CustomerC has 7 sensors: Flush 1/2, Sink 1/2/3, Tap 1/2. A normal flush is ~6 L and a normal sink use is ~1-2 L. For each anomaly listed, explain the likely cause and recommended action.

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 1.2s

Based on the provided data, the top anomalies in the residential CustomerC dataset are: 1. **Anomaly:** Extremely high water consumption by Residential Flush 1 on 2024-06-25 13:13, 2024-05-22 09:41, and 2024-06-27 20:32, with consumption values of 149.61 L, 200.65 L, and 219.33 L, respectively. **Likely cause:** These high consumption values are likely due to a malfunctioning or clogged flush mechanism, causing the toilet to use significantly more water than usual. **Recommended action:** Inspect the toilet's flush mechanism and clear any blockages. If the issue persists, consider replacing the toilet's flapper or fill valve. 2. **Anomaly:** Extremely high water consumption by Residential Sink 3 on 2024-08-30 08:24, with a consumption value of 82.84 L. **Likely cause:** This high consumption value is likely due to a leaky faucet or a malfunctioning aerator, causing the sink to use significantly more water than usual. **Recommended action:** Inspect the faucet and aerator for any signs of leaks or damage. Replace the aerator and tighten any loose connections. 3. **Anomaly:** Extremely high water consumption by Residential Flush 1 on 2024-07-10 08:10, with a consumption value of 92.81 L. **Likely cause:** This high consumption value is likely due to a partial clog in the toilet's drain line or a malfunctioning flapper, causing the toilet to use more water than usual. **Recommended action:** Inspect the toilet's drain line and clear any blockages. If the issue persists,... [truncated]

### GPT-OSS 120B (Groq) · 2.1s

## Top Anomalies in **Customer C** (Residential) | # | Date & Time (UTC) | Device (sensor) | **Consumption** (L) | Roll **mean** (L) | **z-score** | How far it is from "normal" | Likely cause | Recommended action |  
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|  
2024-06-25 13:13 | Flush 1 | **149.61 L** | 5.85 L | **14.3** | ≈ 25 × a typical 6 L flush | • Sensor may have stayed "open" after the flush (stuck valve) <br>• Possible pipe burst or major leak in the toilet line | 1. Visually inspect the toilet flush mechanism for a stuck valve or continuous flow. <br>2. Shut off the water to the toilet and re-open to see if flow stops. <br>3. Run a pressure test on the toilet supply line to rule out a burst pipe. <br>4. If the valve is fine, recalibrate/replace the Flush 1 sensor. | 2 | 2024-10-08 08:52 | Flush 1 | **137.01 L** | 5.53 L | **14.02** | ≈ 23 × normal | Same pattern as #1 – repeated high-volume event on the same sensor suggests a persistent fault. | • Check for a slow drip leak in the toilet's fill valve that may have gone unnoticed. <br>• Verify that the sensor's "open/close" detection logic is not being triggered by a vibration or electrical glitch. | 1. Conduct a leak detect test on the toilet (apply food colour to the tank water and watch for colour in the bowl). <br>2. Replace the fill valve if a leak is confirmed. <br>3. Update the sensor... [truncated]

### Qwen3 32B (Groq) · 2.3s

<think> Okay, let's tackle this problem. The user wants me to identify and explain the top anomalies in the CustomerC dataset. They provided data from seven sensors: Flush 1/2 and Sink 1/2/3, Tap 1/2. Normal flush is around 6L, sink use is 1-2L. Each row is a usage event. First, I need to look at the data given. The table has data\_time, device\_label, consumption\_L, roll\_mean, and z\_score. The z-score is a measure of how many standard deviations an element is from the mean. High z-scores indicate anomalies since they're far from the average. Looking at the data, the highest z-scores are for Residential Flush 1. The first entry is 149.61L with a z-score of 14.3. That's way higher than the normal 6L. Similarly, other entries for Flush 1 have consumption around 137-200L, which are all extremely high. Then there's a Sink 3 entry with 14.53L and a z-score of 13.62, which is also a problem since sinks normally use 1-2L. Another Sink 3 entry at 82.84L with a z-score of 13.33 is also a big outlier. So the main anomalies are the high consumption events. For the flushes, using over 100L in one event is way beyond normal. Possible causes could be a toilet leak, like a faulty flapper valve or a fill valve that's stuck open. For the sinks, especially Sink 3, using 14L or 82L is abnormal. That might be due to a dripping faucet, a broken pipe, or someone leaving the tap open accidentally. The user mentioned seven sensors, but the data only shows Flush 1 and Sink 3. Maybe the other senso... [truncated]

### Llama 3.3 70B (Groq) · 1.5s

Based on the provided data, the top anomalies in the residential CustomerC dataset are: 1. **\*\*2024-06-27 20:32: Residential Flush 1 with 219.33 L\*\***: This anomaly has a z-score of 12.86, indicating a significant deviation from the normal flush volume of ~6 L. The likely cause is a faulty or stuck flush valve, causing excessive water flow. Recommended action: Inspect and repair/replace the flush valve to prevent further water waste. 2. **\*\*2024-05-22 09:41: Residential Flush 1 with 200.65 L\*\***: This anomaly has a z-score of 13.36, indicating another instance of excessive water flow. The likely cause is the same as the previous anomaly, a faulty or stuck flush valve. Recommended action: Inspect and repair/replace the flush valve to prevent further water waste. 3. **\*\*2024-08-30 08:24: Residential Sink 3 with 82.84 L\*\***: This anomaly has a z-score of 13.33, indicating a significant deviation from the normal sink use of ~1-2 L. The likely cause is a leaky faucet or an open tap that was not properly closed. Recommended action: Inspect the sink faucet for leaks and ensure that it is properly closed after use. 4. **\*\*2024-06-25 13:13: Residential Flush 1 with 149.61 L\*\***: This anomaly has a z-score of 14.3, indicating another instance of excessive water flow. The likely cause is the same as the previous flush valve anomalies, a faulty or stuck flush valve. Recommended action: Inspect and repair/replace the flush valve to prevent further water waste. 5. **\*\*2025-09-09 13:36: Residential Sink...** [truncated]

### Mistral 7B (Local) · 140.9s

In the provided dataset for CustomerC, several anomalies can be identified based on the high z-scores, which indicate significant deviations from the mean consumption for each device. Here are the top anomalies and their likely causes along with recommended actions: 1. **\*\*Residential Flush 1\*\***: Multiple instances of excessive water usage (z-scores ranging from 12.52 to 14.3) indicate that the user may be using more water than necessary during flushes. This could be due to a running toilet, prolonged flushing, or an issue with the flush mechanism itself. - Recommended action: Investigate the cause of excessive water usage in Residential Flush 1 and address any issues found (e.g., repairing a leaking toilet, adjusting the flush mechanism). 2. **\*\*Residential Sink 3\*\***: Two instances of high water consumption (z-scores of 13.62 and 13.33) suggest that the user may be using more water than usual during sink activities on this particular sink. This could be due to a leaking faucet, prolonged use, or unusual activities like washing large items. - Recommended action: Inspect Residential Sink 3 for any leaks or issues with the faucet and address them if found. If necessary, educate the user about efficient water usage during sink activities. 3. **\*\*Outlier values\*\***: There are two instances where the consumption is significantly higher than the normal range for flushes (200.65 L and 219.33 L). These could be due to a temporary event, such as filling a large container or cleaning a ... [truncated]

### Qwen2.5 3B (Local) · 106.1s

Let's analyze the anomalies in the CustomerC dataset based on the provided data: **### Anomaly 1: \*\*Data Point:\*\* Residential Flush 1, 2024-06-25 13:13, consumption\_L = 149.61 L \*\*Explanation:\*\*** This is a significant outlier with a z-score of 14.3, indicating that the usage (149.61 L) is far above the mean (roll\_mean = 5.85 L). A normal flush should be around 6 L. **\*\*Likely Cause:\*\*** There could be an error in data collection or reporting for this event. It might also indicate a malfunctioning sensor, such as a leak or faulty device that has been running continuously. **\*\*Recommended Action:\*\*** Investigate the sensor and associated plumbing to ensure it is not leaking or malfunctioning. If necessary, replace any faulty components and recalibrate the sensor if needed. Consider contacting the property management or homeowner for further investigation. **### Anomaly 2: \*\*Data Point:\*\* Residential Flush 1, 2024-05-22 09:41, consumption\_L = 200.65 L \*\*Explanation:\*\*** This is another outlier with a z-score of 13.36, significantly higher than the mean (roll\_mean = 6.62 L). A normal flush should be around 6 L. **\*\*Likely Cause:\*\*** Similar to the previous anomaly, this could indicate an error in data collection or reporting. It might also suggest that the sensor is malfunctioning and has been running continuously for a long time. **\*\*Recommended Action:\*\*** Investigate the sensor and associated plumbing. Replace any faulty components if necessary. Recalibrate the sensor if needed. Contact the ... [truncated]

### [S4] Medium

*Which customer has the highest average daily water consumption? Rank all 4 customers and explain the difference given their profiles: Gym (showers), CustomerA (office toilet bloc), CustomerB (sanitary bloc with wudu sink), CustomerC (residential home).*

✓ Real sensor data injected into this prompt

### Llama 3.1 8B (Groq) · 0.8s

Based on the provided data, the ranking of the customers by their average daily water consumption is as follows: 1. **\*\*CustomerB\*\***: 1003.4 L/day 2. **\*\*CustomerA\*\***: 604.1 L/day 3. **\*\*Gym\*\***: 218.1 L/day 4. **\*\*CustomerC\*\***: 258.1 L/day However, since CustomerC's average daily consumption is higher than the Gym's, CustomerC should be ranked second, and the Gym should be ranked third. The difference in average daily water consumption between CustomerB and CustomerA is 399.3 L/day. This is likely due to the fact that CustomerB has a sanitary bloc with a wudu sink, which requires more water for cleaning and flushing. CustomerA, on the other hand, has an office toilet bloc, which typically uses less water. The difference between CustomerC and the Gym is 40 L/day, which is relatively small. This suggests that both CustomerC's residential home and the Gym have similar water usage patterns, possibly due to the presence of showers in both locations.

### GPT-OSS 120B (Groq) · 2.1s

**\*\*Average daily water consumption (L / day)\*\* | Rank | Customer | Total volume (L) | Monitoring period (days) | Avg. L / day\* |**  
 |-----|-----|-----|-----|-----| | 1 | **\*\*Customer B\*\*** (sanitary bloc with wudu sink) | 214 731.3 L | 214 days | **\*\*1 003.4 L / day\*\*** | | 2 | **\*\*Customer A\*\*** (office toilet bloc) | 105 712.5 L | 175 days | **\*\*604.1 L / day\*\*** | | 3 | **\*\*Customer C\*\*** (residential home) | 162 082.1 L | 628 days | **\*\*258.1 L / day\*\*** | | 4 | **\*\*Gym\*\*** (showers) | 162 679.6 L | 746 days | **\*\*218.1 L / day\*\*** |  
 \\*Average daily consumption is calculated as **\*\*total volume ÷ number of days\*\*** – the values shown are exactly those supplied in the data set. --- **### Why the differences?** | Customer | Typical water■using fixtures & activity | How that drives the daily average |  
 |-----|-----|-----|-----| | **\*\*Customer B – sanitary bloc with wudu sink\*\*** | Multiple



wash■basins, ablution (wudu) sink, possibly separate male/female sections, high■flow taps for ritual washing. | Ritual washing often involves repeated, relatively long flows (e.g., 2 L / s for 30 s per person). Combined with a high foot■traffic volume (students, worshippers), the per■person water draw is large, pushing the daily average above \*\*1 m³\*\*. | | \*\*Customer A – office toilet bloc\*\* | Standard toilets, hand■wash basins, occasional urinals. | Office use is steady throughout the workday but limited to short flu... [truncated]

### Qwen3 32B (Groq) · 2.1s

<think> Okay, let's tackle this question. The user wants to know which customer has the highest average daily water consumption and to rank all four customers, explaining the differences based on their profiles. First, I need to look at the data provided. The data shows each customer's average daily consumption and the total over their respective days. The customers are Gym, CustomerA, CustomerB, and CustomerC. Their profiles are given as Gym (showers), CustomerA (office toilet bloc), CustomerB (sanitary bloc with wudu sink), and CustomerC (residential home). The average daily consumption values are: - CustomerB: 1003.4 L/day - CustomerA: 604.1 L/day - Gym: 218.1 L/day - CustomerC: 258.1 L/day So, the highest is CustomerB, followed by CustomerA, then CustomerC, and finally the Gym. Wait, but CustomerC's average is higher than the Gym's. Let me check again. CustomerC is 258.1 L/day, which is more than the Gym's 218.1. So the order should be CustomerB, CustomerA, CustomerC, Gym. That makes sense because CustomerB is a sanitary bloc with a wudu sink, which might involve more water usage per event compared to the others. The Gym uses showers, which might be shorter and less frequent than the sanitary bloc. CustomerA is an office toilet bloc, which would have regular use but maybe not as high as a sanitary bloc. CustomerC is a residential home, which might have varied usage but lower than commercial facilities. I need to make sure I'm not confusing total consumption with avera... [truncated]

### Llama 3.3 70B (Groq) · 2.5s

To determine the customer with the highest average daily water consumption, we can directly compare the provided average daily consumption values for each customer. The ranking from highest to lowest average daily water consumption is: 1. CustomerB: 1003.4 L/day 2. CustomerA: 604.1 L/day 3. CustomerC: 258.1 L/day 4. Gym: 218.1 L/day The difference in consumption can be explained by the profiles of each customer: - CustomerB has the highest consumption, likely due to the sanitary bloc with a wudu sink, which may require more water for cleaning and ritual washing. - CustomerA has an office toilet bloc, which would have a moderate number of users and therefore moderate water consumption. - CustomerC, a residential home, has a lower consumption, possibly due to fewer occupants or more water-efficient appliances. - The Gym, with showers, has the lowest consumption among the four, possibly due to the use of low-flow showerheads or limited hours of operation.

### Mistral 7B (Local) · 117.0s

To find the average daily water consumption for each customer, we first need to calculate their total water usage in a day based on the provided data. 1. CustomerA: Total water usage is given as 105,712.5 L over 175 days. So, the average daily water consumption is  $105,712.5 \text{ L} / 175 \text{ days} = 604.1 \text{ L/day}$  (already provided in the data). 2. CustomerB: Total water usage is given as 214,731.3 L over 214 days. So, the average daily water consumption is  $214,731.3 \text{ L} / 214 \text{ days} = 1003.4 \text{ L/day}$  (already provided in the data). 3. CustomerC: Total water usage is given as 162,082.1 L over 628 days. So, the average daily water consumption is  $162,082.1 \text{ L} / 628 \text{ days} = 258.1 \text{ L/day}$  (already provided in the data). 4. Gym: Total water usage is given as 162,679.6 L over 746 days. So, the average daily water consumption is  $162,679.6 \text{ L} / 746 \text{ days} = 218.1 \text{ L/day}$  (already provided in the data). Ranking the customers from highest to lowest average daily water consumption: 1. CustomerB: 1003.4 L/day 2. Gym: 218.1 L/day 3. CustomerA: 604.1 L/day 4. CustomerC: 258.1 L/day The difference in water consumption among the customers is significant. CustomerB has the highest average daily water consumption, which can be attributed to the presence of a wudu sink that likely requires more water for ablution purposes compared to the other locations. The gym and CustomerA have similar but lower water consumption levels, possibly due to showers and office toilet usage respectively. CustomerC, being a residential ho... [truncated]

### Qwen2.5 3B (Local) · 72.6s

To determine which customer has the highest average daily water consumption, we need to calculate the average daily usage for each customer by dividing their total water usage by the number of days they have been monitored. \*\*CustomerA (Office Toilet Bloc):\*\* - Total Water Usage: 105,712.5 L - Number of Days Monitored: 175 days - Average Daily Consumption =  $\frac{105,712.5}{175} \approx 603.51 \text{ L/day}$  \*\*CustomerB (Sanitary Bloc with Wudu Sink):\*\* - Total Water Usage: 214,731.3 L - Number of Days Monitored: 214 days - Average Daily Consumption =  $\frac{214,731.3}{214} \approx 1002.50 \text{ L/day}$  \*\*CustomerC (Residential Home):\*\* - Total Water Usage: 162,082.1 L - Number of Days Monitored: 628 days - Average Daily Consumption =  $\frac{162,082.1}{628} \approx 259.47 \text{ L/day}$  \*\*Gym (Showers):\*\* - Total Water Usage: 162,679.6 L - Number of Days Monitored: 746 days - Average Daily Consumption =  $\frac{162,679.6}{746} \approx 218.03 \text{ L/day}$  \*\*Ranking from highest to lowest average daily consumption:\*\* 1. \*\*CustomerB (Sanitary Bloc with Wudu Sink):\*\* 1002.50 L/day 2. \*\*CustomerA (Office Toilet Bloc):\*\* 603.51 L/day 3. \*\*CustomerC (Residential Home):\*\* 259.47 L/day 4. \*\*Gym (Showers):\*\* 218.03 L/day \*\*Explanation of Differences:\*\* - \*\*CustomerB:\*\* The highest average daily consumption is due to the combination of a sanitary bloc with a wudu sink, which typically includes multiple water-using fixtures such as faucets and taps. This profile suggests that customers in this ... [truncated]

**[S5] Medium**

Compare CustomerA and CustomerB water consumption side by side. CustomerA is an office toilet bloc with 4 aggregated bloc sensors. CustomerB is a sanitary bloc (2 WCs, sinks, wudu sink) with 1 aggregated sensor. Explain the difference in total and daily consumption. Which is more water-efficient per...

✓ Real sensor data injected into this prompt

**Llama 3.1 8B (Groq) · 0.7s**

To compare CustomerA and CustomerB water consumption, we need to calculate the total consumption and daily consumption for each customer. **\*\*Total Consumption:\*\*** CustomerA: 105,712.5 L CustomerB: 214,731.3 L CustomerB has a significantly higher total consumption than CustomerA, with a difference of 109,018.8 L. **\*\*Daily Consumption:\*\*** To calculate the daily consumption, we need to know the duration of the monitoring period for each customer. CustomerA: 2025-05-31 → 2025-11-22 = 176 days CustomerB: 2025-10-06 → 2026-05-09 = 220 days Daily Consumption (CustomerA): 105,712.5 L / 176 days ≈ 601.1 L/day Daily Consumption (CustomerB): 214,731.3 L / 220 days ≈ 975.3 L/day **\*\*Water Efficiency per User:\*\*** To compare the water efficiency per user, we need to know the number of users for each customer. CustomerA: 4 aggregated bloc sensors (assuming 1 sensor per user) CustomerB: 1 aggregated sensor (assuming 1 sensor per user, but with 2 WCs and other fixtures) However, we cannot accurately determine the number of users for CustomerB based on the provided data. Assuming both customers have a similar number of users, we can compare their daily consumption per sensor. CustomerA: 601.1 L/day / 4 sensors ≈ 150.3 L/day/sensor CustomerB: 975.3 L/day / 1 sensor ≈ 975.3 L/day/sensor Based on this comparison, CustomerA appears to be more water-efficient per user, with a daily consumption of approximately 150.3 L/day/sensor, compared to CustomerB's 975.3 L/day/sensor. However, please ... [truncated]

**GPT-OSS 120B (Groq) · 2.0s**

**\*\*Water Use summary (as recorded by the aggregated sensors)\*\*** | Customer A – office toilet bloc | Customer B – sanitary bloc |  
 |-----|-----|-----| Total volume measured | **\*\*105 712.5 L\*\*** | **\*\*214 731.3 L\*\*** |  
 Monitoring period | 31 May 2025 → 22 Nov 2025 | 06 Oct 2025 → 09 May 2026\* | Number of days (≈) | **\*\*176 days\*\*** | **\*\*226 days\*\*** |  
 Avg. daily consumption | **\*\*105 712.5 L ÷ 176 d ≈ 600.6 L / day\*\*** | **\*\*214 731.3 L ÷ 226 d**

**Qwen3 32B (Groq) · 2.9s**

<think> Okay, let's tackle this comparison between CustomerA and CustomerB. First, I need to understand the data provided. CustomerA is an office toilet block with 4 sensors, and CustomerB is a sanitary block with 1 sensor covering 2 WCs, sinks, and a wudu sink. The time periods are different: CustomerA from May 31, 2025, to November 22, 2025, and CustomerB from October 6, 2025, to May 9, 2026. The total consumption is 105,712.5 L for CustomerA and 214,731.3 L for CustomerB. First, I should calculate the total consumption for each. CustomerA used about 105,712.5 liters over roughly 6 months, while CustomerB used over 214,700 liters over about 7 months. But the time periods aren't the same length. CustomerA's period is 6 months (May to November), and CustomerB's is 7 months (October to May). To compare daily consumption, I need to convert the total liters to daily averages. For CustomerA: 105,712.5 L divided by the number of days from May 31 to November 22. Let me calculate that. May has 31 days, but starting on the 31st, so May is 1 day. June has 30, July 31, August 31, September 30, October 31, November 22. Total days: 1 + 30 + 31 + 31 + 30 + 31 + 22 = 175 days. Wait, that can't be right. Wait, May 31 to November 22 is 6 months and 22 days. Let me count properly. From May 31 to November 30 is 6 months. May 31 to June 30 is 30 days (since May has 31, starting on the 31st, so June 1 is day 1). Then June 1 to June 30 is 30 days, July 31, August 31, September 30, October 31, N... [truncated]

**Llama 3.3 70B (Groq) · 0.8s**

To compare the water consumption of CustomerA and CustomerB, we need to calculate the total consumption and daily consumption for each customer. CustomerA: - Total consumption: 105,712.5 L - Duration: 2025-05-31 to 2025-11-22, which is approximately 206 days (from May 31 to November 22) - Daily consumption: 105,712.5 L / 206 days ≈ 513.1 L/day CustomerB: - Total consumption: 214,731.3 L - Duration: 2025-10-06 to 2026-05-09, which is approximately 216 days (from October 6 to May 9) - Daily consumption: 214,731.3 L / 216 days ≈ 994.5 L/day CustomerA has a lower daily consumption (513.1 L/day) compared to CustomerB (994.5 L/day). However, to determine which is more water-efficient per user, we need more information about the number of users for each customer. Since the number of users is not provided, we cannot make a direct comparison of water efficiency per user.

**Mistral 7B (Local) · 96.4s**

To compare the water consumption of CustomerA (office toilet bloc) and CustomerB (sanitary bloc), let's first calculate the total water usage for each customer, and then determine their daily consumption. Customer A: Total Water Usage = 105,712.5 L Duration = 184 days (from May 31, 2025 to November 22, 2025) Daily Consumption = Total Water Usage / Duration Daily Consumption = 105,712.5 L / 184 days ≈ 576.39 L/day Customer B: Total Water Usage = 214,731.3 L Duration = 262 days (from October 6, 2025 to May 9, 2026) Daily Consumption = Total Water Usage / Duration Daily Consumption = 214,731.3 L / 262 days ≈ 817.2 L/day Comparing the daily consumption of both customers: - Customer A has a daily water consumption of approximately 576.39 L/day. - Customer B has a daily water consumption of approximately 817.2 L/day. To determine which is more water-efficient per user, we need to know the number of users for each customer. However, since that information is not provided in the data, I cannot definitively say which is more water-efficient per user. But based on the total and daily consumption numbers, Customer A appears to be using less water compared to Customer B.

**Qwen2.5 3B (Local) · 69.7s**

To analyze the water consumption and efficiency between CustomerA (the office toilet bloc) and CustomerB (the sanitary bloc), we need to calculate their total and daily average water usage, as well as compare them. **### Total Water Usage: \*\*CustomerA\*\*:** - Total Consumption = 105,712.5 L **\*\*CustomerB\*\*:** - Total Consumption = 214,731.3 L **\*\*Difference in Total Consumption\*\*:** \[

$\text{Difference} = 214,731.3 - 105,712.5 = 109,018.8 \text{ L}$  CustomerB used significantly more water than CustomerA over the given period.

**### Daily Average Water Usage:** To calculate daily average consumption for both customers, we need to know the total duration of their usage periods in days.

- **Duration for CustomerA:** 2025-11-22 - 2025-05-31 = 198 days
- **Duration for CustomerB:** 2026-05-09 - 2025-10-06 = 174 days

**CustomerA's Daily Average Consumption:**  $\frac{105,712.5}{198} \approx 532.4 \text{ L/day}$

**CustomerB's Daily Average Consumption:**  $\frac{214,731.3}{174} \approx 1236.0 \text{ L/day}$

**### Water-Efficiency Comparison:** To determine which customer is more water-efficient per user, we need to know the number of users for each customer.

- **CustomerA:** Assuming one user (the office toilet bloc).
- **CustomerB:** 2 WCs + 1 sink = 3 users.

**Water Efficiency Calculation:**

- **CustomerA's Water-Efficiency:**  $\frac{532.4}{1} = 532.4 \text{ L/user/day}$
- **CustomerB's Water-Efficiency:**  $\frac{1236.0}{3} = 412.0 \text{ L/user/day}$  [truncated]

5 Summary Scorecard

| Model                | Avg Score | Avg Time | Errors | Monthly \$ |
|----------------------|-----------|----------|--------|------------|
| Llama 3.1 8B (Groq)  | 0.000     | 1.3s     | 0      | \$0.15     |
| GPT-OSS 120B (Groq)  | 0.000     | 2.1s     | 0      | \$0.42     |
| Qwen3 32B (Groq)     | 0.000     | 2.9s     | 0      | \$1.01     |
| Llama 3.3 70B (Groq) | 0.000     | 1.8s     | 0      | \$1.63     |
| Mistral 7B (Local)   | 0.000     | 121.9s   | 0      | \$0.00     |
| Qwen2.5 3B (Local)   | 0.000     | 71.9s    | 0      | \$0.00     |

6 Cost–Quality Recommendation

Budget score = quality ÷ (1 + log(1 + monthly cost)). Higher = better quality per dollar.

|   | Model                | Avg Quality | Monthly \$ | Budget Score |
|---|----------------------|-------------|------------|--------------|
| ★ | Llama 3.1 8B (Groq)  | 0.000       | \$0.15     | 0.000        |
| 2 | GPT-OSS 120B (Groq)  | 0.000       | \$0.42     | 0.000        |
| 3 | Qwen3 32B (Groq)     | 0.000       | \$1.01     | 0.000        |
|   | Llama 3.3 70B (Groq) | 0.000       | \$1.63     | 0.000        |
|   | Mistral 7B (Local)   | 0.000       | \$0.00     | 0.000        |
|   | Qwen2.5 3B (Local)   | 0.000       | \$0.00     | 0.000        |

★ Recommendation: Llama 3.1 8B (Groq) offers the best quality-to-cost balance. For on-premises deployments, prefer the highest-scoring local model.

7 Proposed Production Fallback Chain

- 1 Llama 3.1 8B (Groq) — Best budget score
- 2 GPT-OSS 120B (Groq) — Rate-limit fallback
- 3 Qwen3 32B (Groq) — Rate-limit fallback
- 4 Llama 3.3 70B (Groq) — Rate-limit fallback
- 5 Mistral 7B (Local) — Local fallback — zero cost / data privacy
- 6 Qwen2.5 3B (Local) — Local fallback — zero cost / data privacy

Groq keys rotate automatically across KEY\_1–KEY\_4. Reorder \_PROVIDERS in llm\_client.py to change the agent default.